

VARITHON 2026 – Stage 1 Problem Discovery Report (PDR)

Team ID: V26-GF-266

Track: Open Innovation (Software Based)

Project Title: VARI OS – AI Decision Intelligence & Simulation Platform
for Mass Gathering Management

Team Leader: Disha Gudup

Team Members: Bhavikesh Hedau, Anuj Gardi, Harshal Ramteke

1. Problem Discovery

Problem Statement

Every year, millions of Varkaris participate in the Vari pilgrimage. Although CCTV, weather updates, volunteers, GPS, medical camps and police control rooms generate valuable information, officials still coordinate largely through manual communication. During emergencies, time is lost verifying incidents, identifying available resources and deciding the best response. The core problem is decision latency—not lack of data. Delayed decisions increase the risk of crowd congestion, heat-related illness, medical emergencies and inefficient resource allocation.

Evidence & Need

- One of the world's largest recurring pilgrimages involving millions of pilgrims.
- Multiple agencies coordinate simultaneously with fragmented information.
- Heat, crowd density, lost pilgrims and traffic require rapid decisions.
- Current systems monitor events but do not proactively recommend actions.



Img1 :Heat Exhaustion / Pilgrims Resting Due to Fatigue

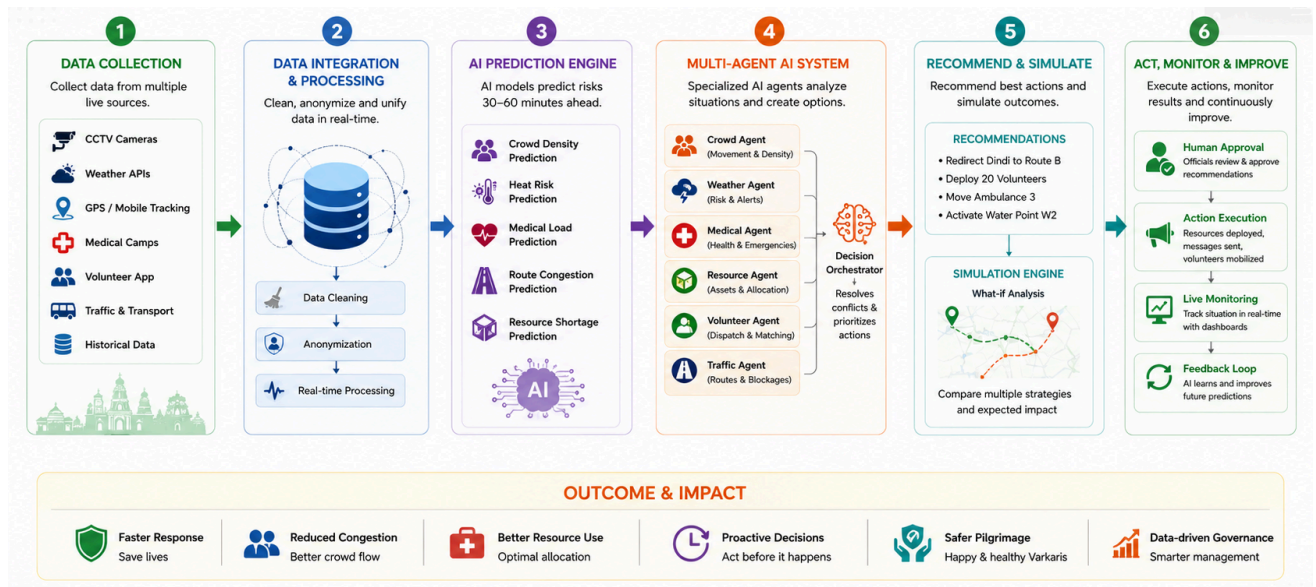
img2 :Ambulance or Medical Team Struggling Through Crowds

2. Proposed Solution

VARI OS

VARI OS is an AI-powered Decision Intelligence & Simulation Platform that transforms multiple live data sources into actionable recommendations. Rather than acting as another dashboard, it predicts future risks, recommends resource deployment and simulates outcomes before officials execute decisions

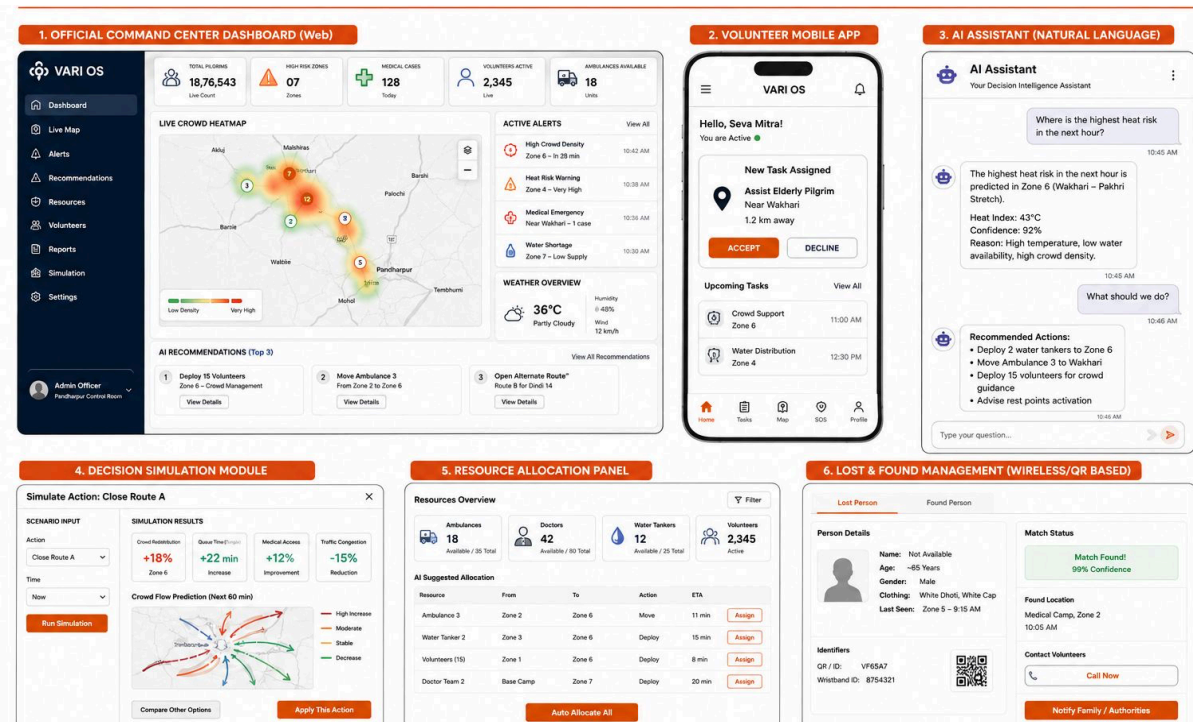
Core Workflow



Key Features

- Predict crowd congestion, heat risk and medical overload 30–60 minutes in advance.
- Recommend deployment of ambulances, volunteers, water points and police.
- Natural-language command center for administrators.
- Decision simulation to compare response strategies.
- Explainable AI with confidence score and reasoning.

Wireframe



3. Technology, Cost & Innovation

Technology Stack

Frontend: React + Tailwind CSS

Backend: FastAPI

AI Orchestration: LangGraph

LLM: Gemini/OpenAI/Groq (prototype)

Database: PostgreSQL

Maps: Leaflet/OpenStreetMap

Analytics: Python, Pandas

Deployment: Docker

Estimated Prototype Cost

Item	Estimated Cost (INR)
Cloud APIs & AI credits	8,000
Hosting & Database	3,000
Development & Testing	12,000
Miscellaneous	2,000
Total Prototype	25,000

The prototype can be developed at minimal cost by leveraging existing infrastructure, open-source technologies, and free-tier cloud services.

Innovation

Unlike traditional dashboards, VARI OS predicts risks, recommends actions and simulates operational outcomes before execution, enabling proactive governance rather than reactive monitoring.

4. Impact & Future Scope

Expected Impact

- Faster emergency response
- Improved crowd safety
- Better volunteer coordination
- Optimized use of ambulances, water and medical teams

- Reduced decision-making time
- Safer experience for elderly and vulnerable pilgrims

Future Scope

Integration with live CCTV analytics, IoT sensors, offline mesh communication, digital twin simulation, QR-based pilgrim assistance and deployment for Kumbh Mela, Ganesh Visarjan, marathons and disaster relief operations.

Conclusion

VARI OS is designed as an AI-powered operational intelligence platform that enables authorities to anticipate risks, evaluate response strategies and deploy resources proactively. By combining predictive analytics, explainable AI and simulation-based decision support, the platform enhances public safety while remaining practical, scalable and cost-effective for the Vari pilgrimage and other large-scale public gatherings.